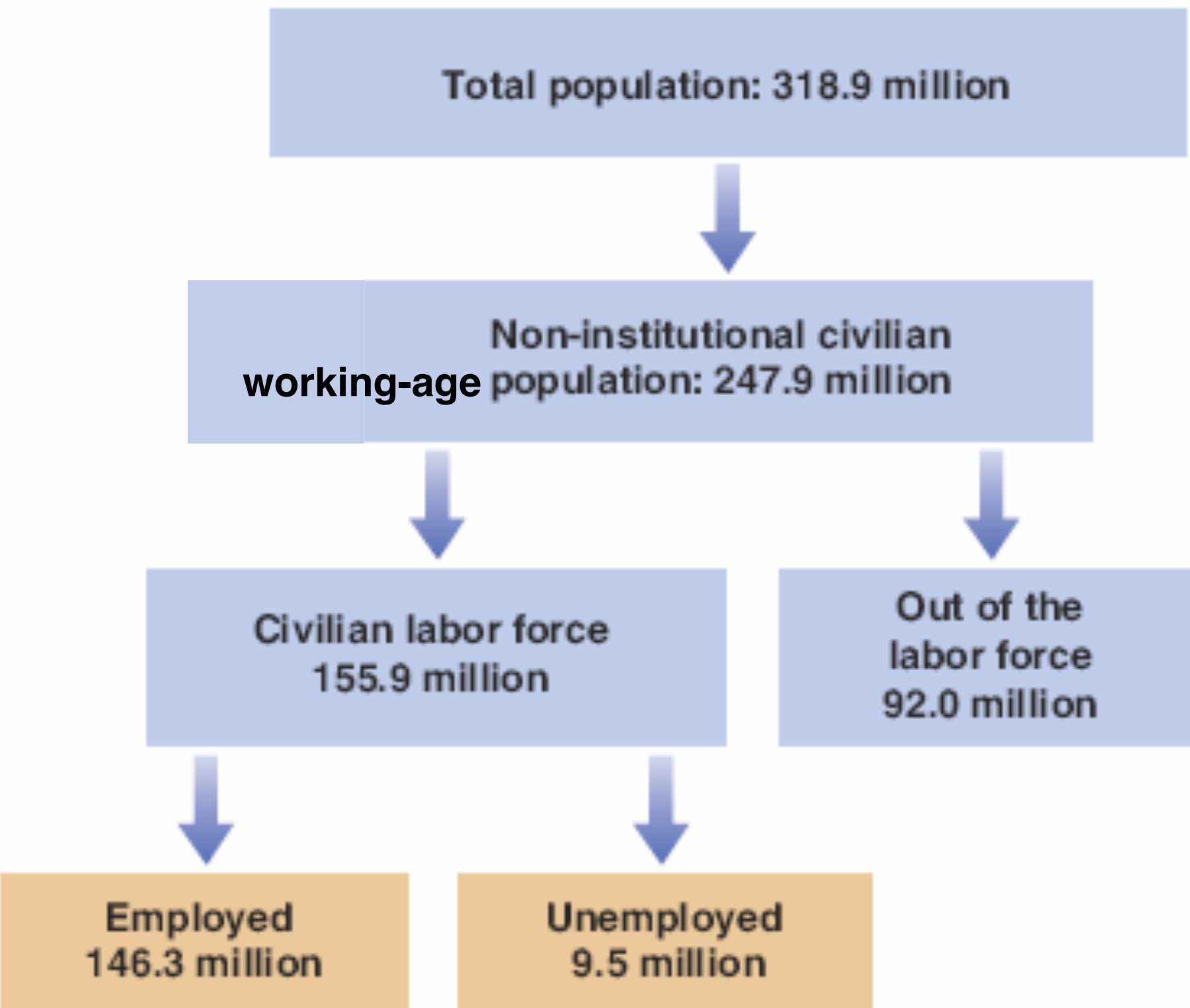


INTERMEDIATE MACROECONOMICS
MACROECONOMIC CONCEPTS
6. UNEMPLOYMENT

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US LABOR MARKET IN 2014



- measured using large surveys of households
- in the US: Current Population Survey(CPS) based on responses of 60,000 households every month since 1940
- CPS is conducted by BLS

WORKING-AGE POPULATION

- a person is in the working-age population if
 - she is 16 years of age and older (no upper age limit)
 - she resides in the 50 states and the District of Columbia
 - she is not inmate of institutions (penal and mental facilities, homes for the aged)
 - she is not on active duty in the Armed Forces

EMPLOYMENT / UNEMPLOYMENT

- a person is employed if she has a full-time job or does temporary or part-time work during the survey week
- a person is unemployed if
 - she does not have a job
 - she is available to work
 - she has been actively looking for a job in the last four weeks (contacting firms, sending CVs, applying to vacancies)

UNEMPLOYMENT RATE

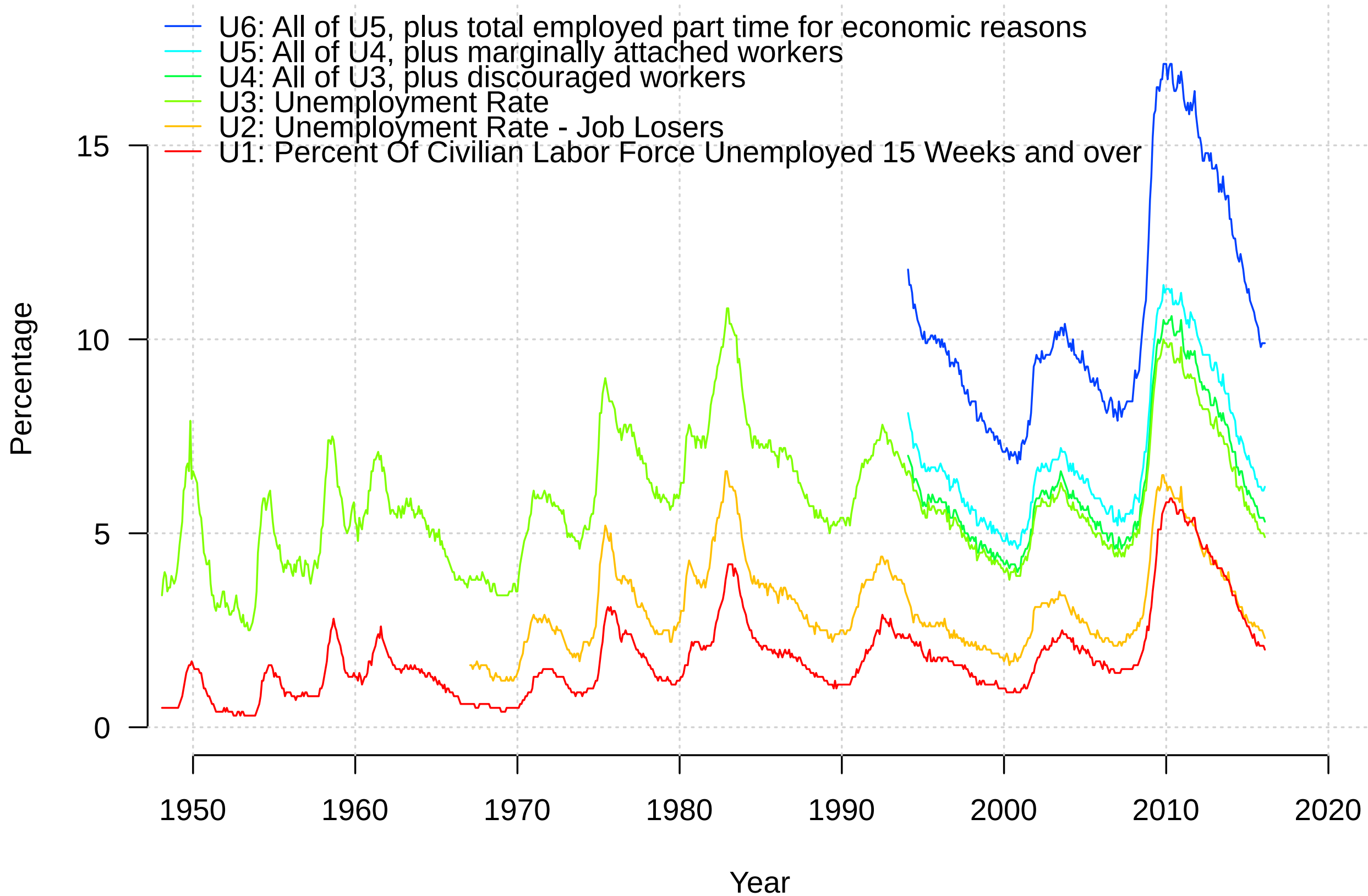
$$\text{unemployment rate} = \frac{\text{unemployment}}{\text{labor force}}$$

- unemployment rate in the US in 2014:
 - $9.5 / 155.9 = 6.1\%$
- conversely employment rate = $100\% - 6.1\% = 93.9\%$
 - share of labor force with a job

DISCOURAGED WORKERS

- discouraged workers are those who give up looking for a job
 - they are no longer counted as unemployed
 - but they would like a job if they were offered one
- some measures of unemployment account for them
 - six measures: U1 — U6
 - official measure: U3
- because of discouraged workers, a higher unemployment rate is sometimes associated with a lower participation rate

6 MEASURES OF UNEMPLOYMENT



WHY DO WE CARE ABOUT UNEMPLOYMENT?

- high or low unemployment is a symptom of inefficiency
 - high unemployment: many resources are idle
 - low unemployment: labor shortages so many resources are wasted trying to recruit workers
- unemployment reduces the welfare of the unemployed, especially those unemployed for long periods of time
 - low mental and physical health (anxiety, low self esteem, low life satisfaction), even after controlling for income

EFFECTS OF UNEMPLOYMENT ON HAPPINESS (GERMAN DATA)



IN / OUT OF LABOR FORCE

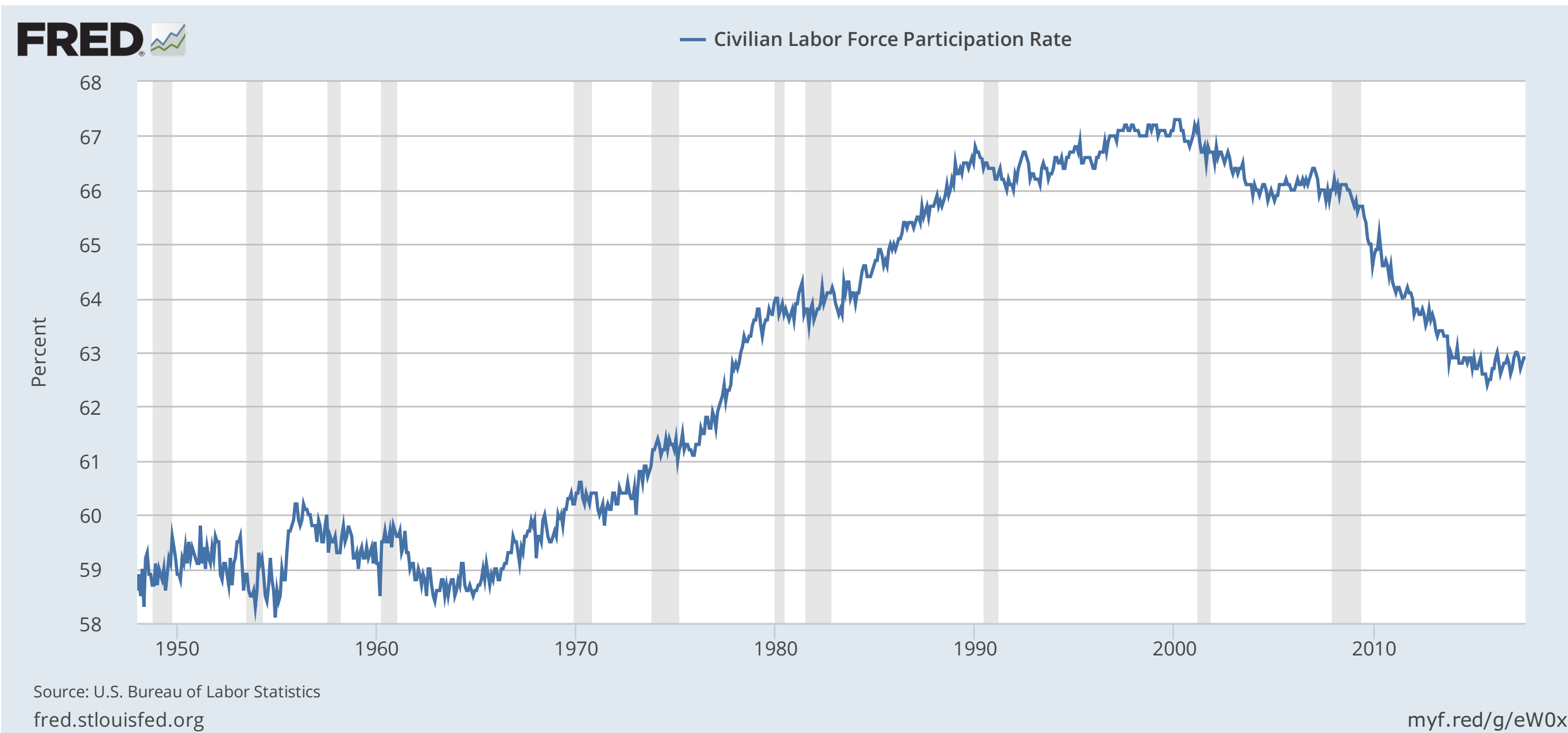
- in the labor force: employment + unemployment
- out of labor force: of working age, but without a job and not looking for a job
 - retired worker
 - university student
 - stay-at-home parent
- working-age population: labor force + out of labor force

LABOR-FORCE PARTICIPATION RATE

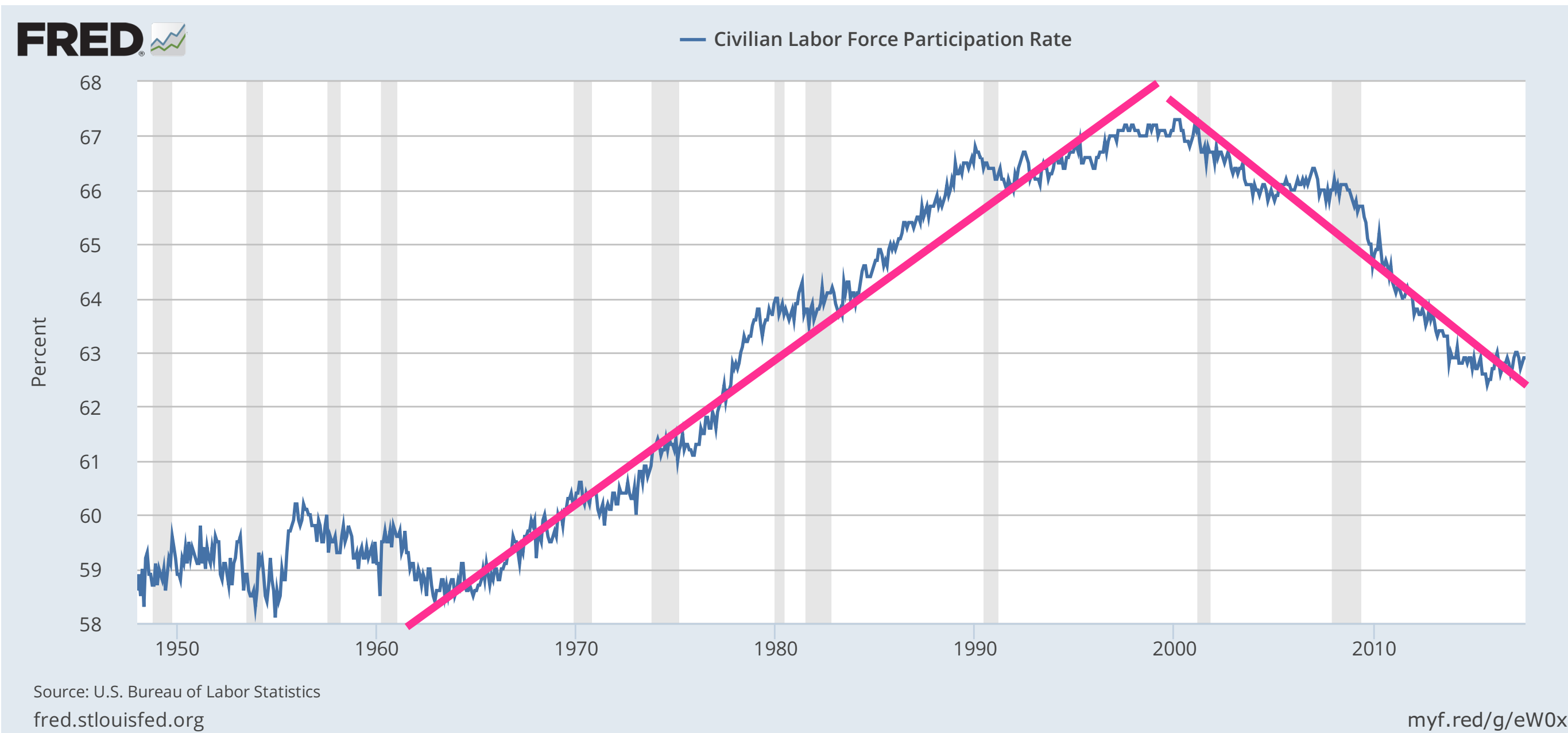
$$\text{participation rate} = \frac{\text{labor force}}{\text{working-age population}}$$

- participation rate in the US in 2014:
 - $155.9 / 247.9 = 62.9\%$
- conversely non-participation rate = $100\% - 62.9\% = 37.1\%$
 - share of working-wage population out of labor force

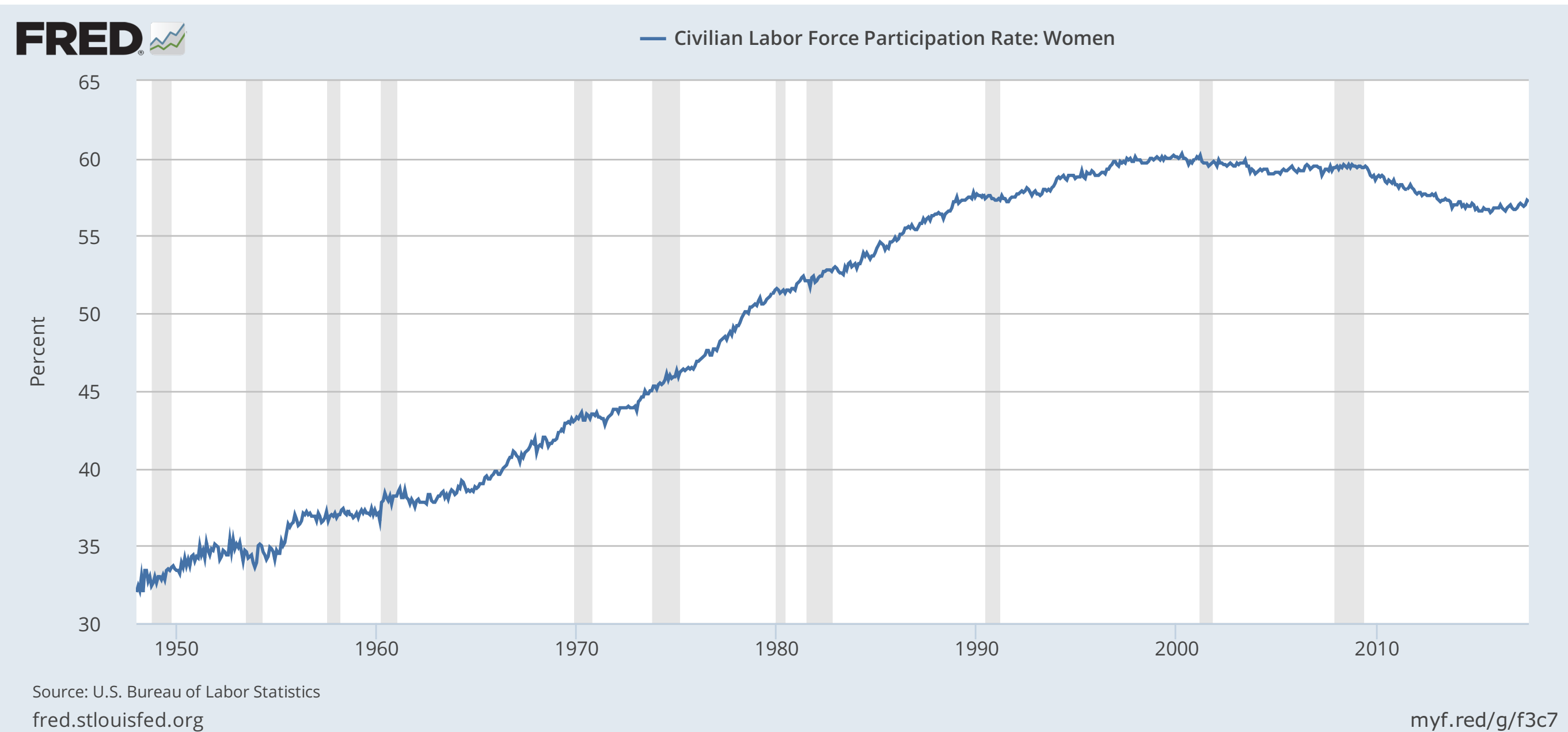
US PARTICIPATION RATE



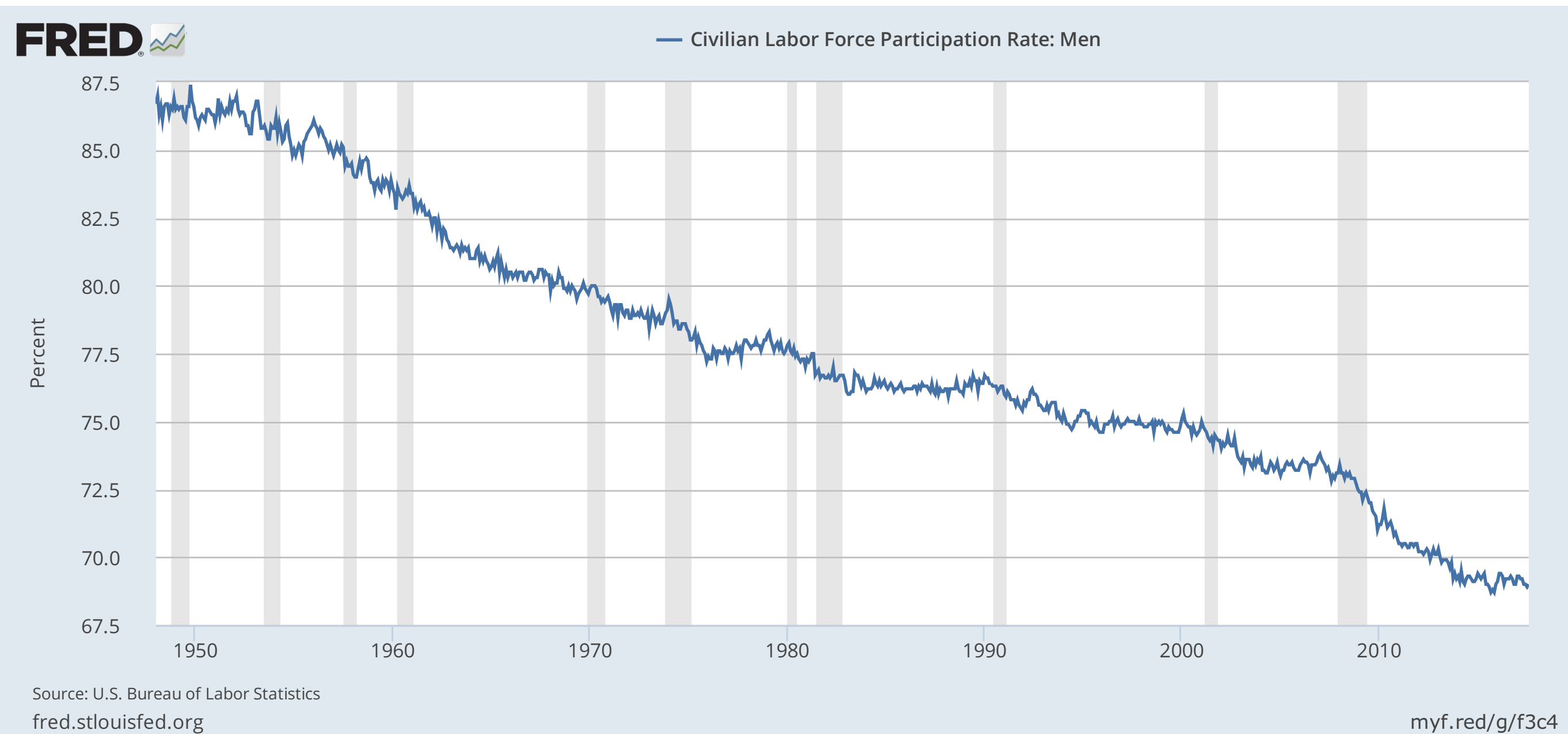
US PARTICIPATION RATE



US PARTICIPATION RATE: WOMEN



US PARTICIPATION RATE: MEN



US LABOR FORCE PARTICIPATION

- <https://www.nytimes.com/2017/09/19/business/economy/labor-shortage.html>
- aging / retirement of baby-boomers contributes to reduced labor-force participation
- 30% of prime-age male who left the labor force are on disability insurance
- opioid prescriptions (painkillers) could explain 20% of decline in male participation and 25% of decline in female participation
- reduced labor-force participation may create labor shortage:
how to attract workers back in labor force?